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This test will give you a broad idea of which topics you most need to spend more time on in your revision.

There is a total of **24** multiple-choice questions, divided into **5** sections.

Test section(s)

- Stoichiometry and Acids/Bases
- Kinetics and Equilibria
- Enthalpy and Entropy
- Inorganic Chemistry
- Organic Chemistry

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1. In a titration, a $25.0\,\mathrm{cm}^3$ aliquot of sulfuric acid ($\mathrm{H_2SO_4}$) reacted with $22.0\,\mathrm{cm}^3$ of $0.10\,\mathrm{mol\,dm^{-3}}$ sodium hydroxide (NaOH) solution. What was the concentration of the sulfuric acid in the aliquot?

- $0.044\,\mathrm{mol\,dm^{-3}}$
- $0.088\,\mathrm{mol\,dm^{-3}}$
- $0.11\,\mathrm{mol\,dm^{-3}}$
- There is not enough information to determine the concentration.

2. Carbon monoxide reacts with oxygen to create carbon dioxide. What volume of oxygen is necessary to fully react with $1.00\,\mathrm{dm}^3$ of carbon monoxide?

- $2.00\,\mathrm{dm}^3$
- $1.00\,\mathrm{dm}^3$
- $0.500\,\mathrm{dm}^3$
- $0.250\,\mathrm{dm}^3$

3. Which of the following is a correct defining description for a Brønsted-Lowry base?

- A Brønsted-Lowry base is a proton acceptor.
- A Brønsted-Lowry base fully dissociates in solution, releasing hydroxide ions.
- A Brønsted-Lowry base is a proton donor.
- A Brønsted-Lowry base is neutralised by accepting electrons.

4. To what concentration would a solution of $0.500\,\mathrm{mol\,dm^{-3}}$ HCl be diluted for the pH to increase by two units?

- A four-fold dilution to $0.125\,\mathrm{mol\,dm^{-3}}$ HCl will increase the pH by two units.

- The pH will decrease when the acid is diluted.
- A hundred-fold dilution to $0.00500\,\mathrm{mol\,dm^{-3}}$ HCl will increase the pH by two units.
- The pH will not change when the acid is diluted.

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Back

5. Which of the following is the correct rate equation for the reaction between sodium hydroxide and aluminium?

$$2\,Al + 2\,NaOH + 6\,H_2O \longrightarrow 2\,Na[Al(OH)_4] + 3\,H_2$$

- $\mathrm{rate} = k[Na[Al(OH)_4]]^2[H_2]^3$
- $\mathrm{rate} = k[Al]^2[NaOH]^2[H_2O]^6$
- There is not enough information to determine the rate equation.
- $\mathrm{rate} = k[Al][NaOH][H_2O]$

6. The rate of a reaction is observed to increase when the temperature is increased. Which of the following does **not** contribute to this?

- The activation energy for the reaction is lower at higher temperatures.
- The particles have higher kinetic energy at higher temperatures.
- The activation energy for the reaction is reached in a higher proportion of collisions at higher temperature.
- The particles collide more frequently at higher temperatures.

7. A catalyst is introduced into a reaction mixture. Which of the following statements correctly describes what happens as a result?

- The concentration of all reagents increases.
- An alternative pathway with a lower activation energy arises.
- The particles have more energy when they collide.
- The equilibrium constant for the reaction increases.

8. Which conditions for the following equilibrium will favour a high yield of the product, $SO_3\,(g)$?

$$2\,SO_2\,(g) + O_2\,(g) \rightleftharpoons 2\,SO_3\,(g) \qquad \Delta H < 0\,kJ\,mol^{-1}$$

- High temperature and high pressure
- High temperature and low pressure
- Low temperature and high pressure
- Low temperature and low pressure

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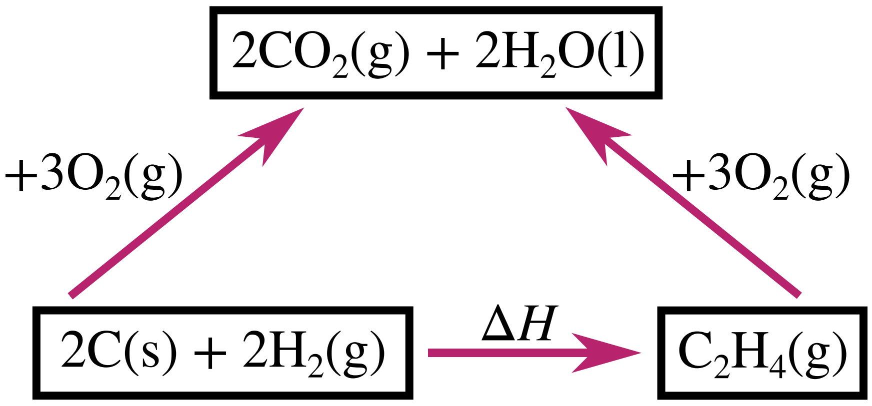
9. Express the enthalpy change for the following reaction in terms of bond enthalpies:

$$C_2H_4\,(g) + H_2\,(g) \longrightarrow C_2H_6\,(g)$$

- BE(C−H)−BE(C−C)−BE(H−H)+BE(C=C)
- BE(H−H)+BE(C=C)−BE(C−H)−BE(C−C)
- BE(H−H)+BE(C=C)−2BE(C−H)−BE(C−C)
- 2BE(C−H)−BE(C−C)−BE(H−H)+BE(C=C)

10. Express the enthalpy change of formation of ethene in terms of the relevant enthalpy changes of combustion.



- $\Delta H = 2\Delta H_c(\mathrm{C}) + 2\Delta H_c(\mathrm{H_2}) + \Delta H_c(\mathrm{C_2H_4})$
- $\Delta H = 3\Delta H_c(\mathrm{C}) + 3\Delta H_c(\mathrm{H_2}) - 3\Delta H_c(\mathrm{C_2H_4})$
- $\Delta H = 3\Delta H_c(\mathrm{C}) + 3\Delta H_c(\mathrm{H_2}) + 3\Delta H_c(\mathrm{C_2H_4})$
- $\Delta H = 2\Delta H_c(\mathrm{C}) + 2\Delta H_c(\mathrm{H_2}) - \Delta H_c(\mathrm{C_2H_4})$

11. If a reaction is described as exothermic, what can we definitely conclude?

- It is fast.
- It is spontaneous at any temperature.

- It is spontaneous, but only at high enough temperatures.
- None of the above

12. Which of the following conclusions about the reaction equation provided is valid?

$$\mathrm{X\,(s) + 2\,Y\,(l) \longrightarrow 2\,Z\,(g)}$$

- The enthalpy change is positive.
- The entropy change is negative.
- The enthalpy change is negative.
- The entropy change is positive.

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Back

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13. Which of the following reactions corresponds to the event relevant for the second ionisation energy of magnesium?

- $Mg^{-}(g) + e^{-} \longrightarrow Mg^{2-}(g)$
- $Mg(g) + 2e^{-} \longrightarrow Mg^{2-}(g)$
- $Mg^{+}(g) \longrightarrow Mg^{2+}(g) + e^{-}$
- $Mg(g) \longrightarrow Mg^{2+}(g) + 2e^{-}$

14. How many bonds does carbon usually form and why?

- 4 because it has 4 valence electrons to share and will have a full octet when forming 4 covalent bonds
- 4 because it needs to lose 4 electrons to have a full outer shell (noble gas configuration)
- 2 because it has 6 valence electrons to share and will have a full octet when forming 2 covalent bonds
- 6 because it needs to lose 6 electrons to have a full outer shell (noble gas configuration)

15. Which entry in the table correctly describes the types of particles present in the different chemical species?

	$HCl(g)$	$HCl(aq)$	$Ne(g)$	$NaCl(s)$

Molecules	Ions	Atoms	Ions

Ions	Ions	Atoms	Molecules

Molecules	Ions	Ions	Molecules

Molecules	Molecules	Atoms	Ions

16. Why does the reactivity of alkali metals generally increase going down the group?

- Going down the group, the nuclear charge increases, so the repulsion felt by the valence electrons is greater and they are easier to lose.
- Going down the group, the nuclear charge increases, so the attraction felt by the incoming electrons is greater and they are easier to gain.
- Going down the group, the incoming electrons feel a smaller repulsion from the nucleus as the higher shells' orbitals are on average further away from the nucleus, so the electrons are easier to gain.
- Going down the group, the valence electrons feel a smaller attraction from the nucleus as the higher shells' orbitals are on average further away from the nucleus, so the electrons are easier to lose.

17. Which of the following explains why transition metals behave differently to main group metals?

- Transition metals have an exactly half-full subshell.
- Transition metals always exist as alloys.
- Transition metals have, or form ions that have, a partially-filled d-subshell.
- Transition metals can use both their s and their p electrons as valence electrons.

18. Which of the following **cannot** act as a ligand in a transition metal complex?

- $\mathrm{H_2O}$
- $\mathrm{OH^-}$
- $\mathrm{NH_4{}^+}$
- $\mathrm{NH_3}$

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isaac You work it out.

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19. Which of the following can be used to easily distinguish between an aldehyde and a ketone, and what is the expected observation?

- Tollens' reagent: a silver mirror is seen for the aldehyde but not the ketone
- Bromine water: decolourisation is observed with the aldehyde but not the ketone
- Silver nitrate: a white precipitate forms with the aldehyde but a yellow precipitate forms with the ketone
- Sodium carbonate: effervescence is seen with the aldehyde but not the ketone

20. Which of the following descriptions of how cyclohexene and benzene react are correct?

- Cyclohexene undergoes electrophilic substitution, benzene undergoes electrophilic addition
- Cyclohexene undergoes electrophilic addition, benzene undergoes electrophilic substitution
- Cyclohexene undergoes nucleophilic addition, benzene undergoes nucleophilic substitution
- Cyclohexene undergoes nucleophilic substitution, benzene undergoes nucleophilic addition

21. What product do you expect to obtain when ethanal reacts with potassium dichromate?

- ethanone
- ethanol
- ethanal (no reaction)
- ethanoic acid

22. What does a monomer for addition polymerisation usually need to contain?

- an aromatic ring
- a chloride group

- two different functional groups
- a $C{=}C$ double bond

23. An organic compound shows a strong, sharp absorption between $1700\,cm^{-1}$ and $1750\,cm^{-1}$ in its IR spectrum. What could this compound be?

- $CH_2{=}CHCH_2C({=}O)H$
- $CH_2{=}CHCH_2CH_2OH$
- $CH_2{=}CHC{\equiv}CH$
- $CH_2{=}CHCH{=}CH_2$

24. Which of the following is the main product when HBr reacts with 2-methylbut-1-ene?

- 1-bromo-2-methylbutane
- 1,2-dibromo-2-methylbutane
- 2-bromo-2-methylbutane
- 1,2-dibromo-2-methylbut-1-ene

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1. Which of the following is a balanced equation for the reaction of a Group 1 metal with water?

- $2\,M\,(s) + 2\,H_2O\,(l) \longrightarrow 2\,MOH\,(aq) + H_2\,(g)$
- $2\,M\,(s) + H_2O\,(l) \longrightarrow M_2O\,(aq) + H_2\,(g)$
- $M\,(s) + 2\,H_2O\,(l) \longrightarrow M(OH)_2\,(aq) + H_2\,(g)$
- $M\,(s) + H_2O\,(l) \longrightarrow MO\,(aq) + H_2\,(g)$

2. Which of the following statements about strong and weak acids is correct?

- A strong acid can react with both strong and weak bases, while a weak acid can only react with strong bases.
- Whether an acid is strong or weak depends on its concentration.
- When present at the same concentration, a strong acid has a lower pH than a weak acid.
- A strong acid in solution always has a $\mathrm{pH} < 3$.

3. Phenol has a $\mathrm{p}K_\mathrm{a}$ value of 10. Calculate the pH of a $0.10\,\mathrm{mol\,dm^{-3}}$ solution of phenol.

- 1.0
- 5.0
- 5.5
- 10

4. Which of the following statements about buffers is correct?

- Buffers can be produced through partial neutralisation of strong acids with strong bases.
- A buffer is always neutral.

- Buffer solutions resist changes in pH, but will change in pH if subjected to a large enough addition of acid or base.
- A buffer always contains equal concentrations of an acid and its conjugate base.

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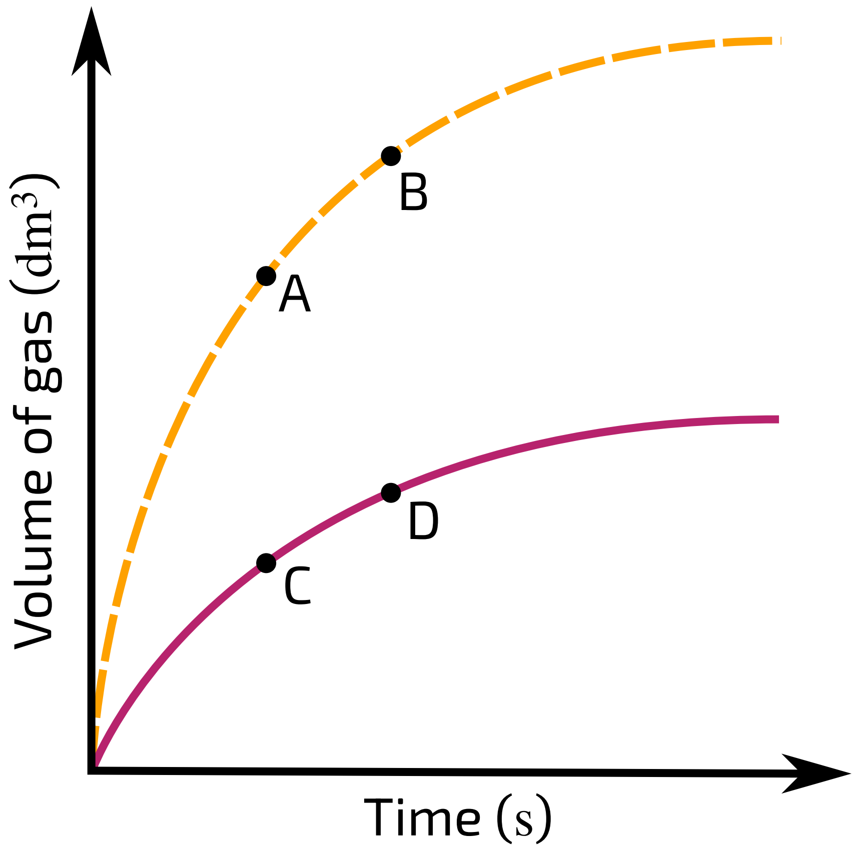
5. At which two points are the rates of the reaction approximately equal, on the basis of the graph provided?



Graph showing volume of gas evolved over time in two different reaction setups.

- A and C
- A and D
- B and C
- B and D

6. The reaction below was experimentally investigated by observing the reaction rate at different concentrations of the reagents X and Y. The results are shown in a table.

$$2\,X\,(aq) + Y\,(aq) \longrightarrow X_2\,Y\,(aq)$$

$[X]/mol\,dm^{-3}$	$[Y]/mol\,dm^{-3}$	$rate/mol\,dm^{-3}\,s^{-1}$
0.10	0.10	0.020
0.30	0.10	0.060

0.10	0.20	0.080

What are the orders of the reaction with respect to X and Y respectively?

	X	Y
○	1	1
○	2	2
○	1	2
○	2	1

7. Two solutions are mixed together and a gas evolves as a reaction takes place. The rate of reaction is seen to decrease over time. Which of the following offers the best explanation for this observation?

- The reaction heats up the mixture, leading to a smaller proportion of collisions being successful.
- The particles have progressively less energy, leading to a lower collision rate.
- The concentration of the reagents decreases over time, leading to a lower collision rate.
- The product blocks the active sites of the reagents, preventing the reaction from going ahead.

8. Write an expression for the equilibrium constant, given the following equilibrium.

$$\mathrm{X}\,(\mathrm{aq}) + \mathrm{Y}\,(\mathrm{aq}) \rightleftharpoons 2\,\mathrm{Z}\,(\mathrm{aq})$$

- $K = \frac{[Z]}{[X][Y]}$
- $K = \frac{[X][Y]}{[Z]}$
- $K = \frac{[Z]^2}{[X][Y]}$
- $K = \frac{[X][Y]}{[Z]^2}$

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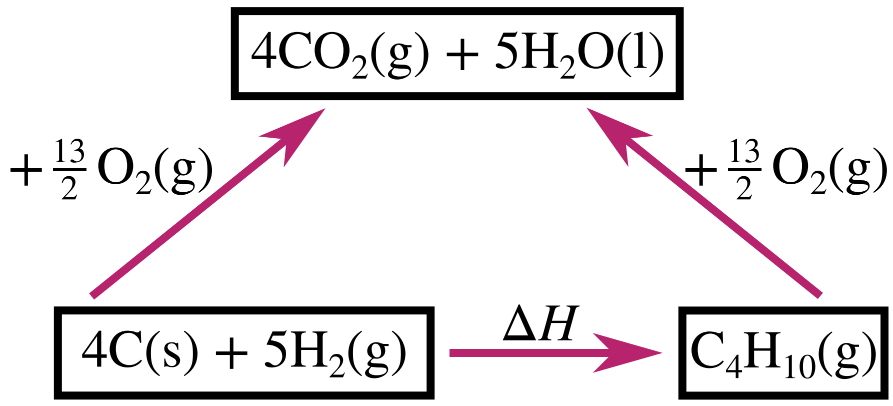
9. Express the enthalpy change of formation of butane in terms of the relevant enthalpy changes of combustion.



- $\Delta H = 4\Delta H_c(C) + 5\Delta H_c(H_2) - \frac{13}{2}\Delta H_c(C_4H_{10})$
- $\Delta H = 4\Delta H_c(C) + 5\Delta H_c(H_2) - \Delta H_c(C_4H_{10})$
- $\Delta H = \Delta H_c(C_4H_{10}) - 4\Delta H_c(C) - 5\Delta H_c(H_2)$
- $\Delta H = \frac{13}{2}\Delta H_c(C_4H_{10}) - 4\Delta H_c(C) - 5\Delta H_c(H_2)$

10. What does a positive enthalpy change indicate about a reaction?

- It cannot go ahead.
- It can go ahead only in the presence of a catalyst.
- It is slow.
- None of the above

11. Which of the following statements about lattice enthalpies of formation is correct?

- The lattice enthalpy of formation does not depend on the charges of the ions.
- The lattice enthalpy of formation becomes more negative the higher in value the charges on the ions are.
- The lattice enthalpy of formation does not depend on the size of the ions.

- [ ] The lattice enthalpy of formation becomes more negative the larger the ions are.

12. Which of the following reactions will have an enthalpy change corresponding to the lattice enthalpy of formation for magnesium chloride?

- [ ] $Mg^{+}(g) + Cl^{-}(g) \longrightarrow MgCl(s)$
- [ ] $Mg^{2+}(g) + 2\,Cl^{-}(g) \longrightarrow MgCl_2(s)$
- [ ] $Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$
- [ ] $Mg(s) + \frac{1}{2}Cl_2(g) \longrightarrow MgCl(s)$

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Back

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13. A given number of moles of $\mathrm{Mg(OH)_2}$ is added to $1.0\,\mathrm{dm^3}$ of water and the pH is recorded. The same number of moles of $\mathrm{Ba(OH)_2}$ is added to another beaker containing $1.0\,\mathrm{dm^3}$ of water and the pH is found to be higher. Explain this observation.

- $\mathrm{Ba(OH)_2}$ contains more hydroxide ions per mole than $\mathrm{Mg(OH)_2}$
- $\mathrm{Ba(OH)_2}$ is more soluble than $\mathrm{Mg(OH)_2}$
- $\mathrm{Mg(OH)_2}$ is more soluble than $\mathrm{Ba(OH)_2}$
- $\mathrm{Mg(OH)_2}$ contains more hydroxide ions per mole than $\mathrm{Ba(OH)_2}$

14. Which of the following correctly describes what happens to the first ionisation energy when crossing a period in the periodic table?

- Ionisation energy decreases throughout
- Ionisation energy mostly decreases, but there are a few increases.
- Ionisation energy increases throughout.
- Ionisation energy mostly increases, but there are a few decreases.

15. Which of the following best describes what happens when HCl dissolves in water?

- The HCl molecules break apart homolytically in water, giving rise to $\mathrm{H\cdot}$ and $\mathrm{Cl\cdot}$ radicals which interact with the water molecules.
- The $\mathrm{H^+}$ and $\mathrm{Cl^-}$ ions initially arranged in a regular lattice separate and get surrounded by water molecules.
- The HCl molecules become surrounded by water molecules and remain this way.
- The HCl molecules break apart heterolytically in water, giving rise to $\mathrm{H^+}$ and $\mathrm{Cl^-}$ ions which interact with the water molecules.

16. When an excess of HCl is added to a solution containing Cu^{2+} ions, a yellow solution forms. What is the shape of the complex responsible for this colour?

- octahedral
- trigonal planar
- tetrahedral
- square planar

17. What is the oxidation state of Cr in the $[Cr(OH)_6]^{3-}$ complex?

- -3
- 0
- +3
- +6

18. Carbon dioxide sublimates at $-78.5\,^{\circ}C$ while silicon dioxide is a solid at RTP and has a melting point above $1710\,^{\circ}C$. Which of the following best accounts for this difference?

- Both carbon dioxide and silicon dioxide exist as simple covalent molecules, but silicon dioxide has more electrons and therefore forms stronger van der Waals interactions.
- Separating carbon dioxide molecules only requires breaking weak van der Waals interactions, while silicon dioxide exists as an ionic lattice, meaning ionic interactions need to be broken in order for it to melt.
- Separating carbon dioxide molecules only requires breaking weak van der Waals interactions, while silicon dioxide exists as a giant covalent lattice, meaning covalent bonds need to be broken in order for it to melt.
- Both carbon dioxide and silicon dioxide exist as simple covalent molecules, but the bonds in silicon dioxide are stronger.

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19. What is the systematic name of CH_3CH_2CHO?

- ethanone
- propanal
- ethanal
- propanone

20. Which of the following mechanisms takes place when hydroxide ions react with chloromethane?

- electrophilic substitution
- nucleophilic addition
- electrophilic addition
- nucleophilic substitution

21. When phenol reacts with bromine, the expected product is...

- bromobenzene
- 3-bromophenol
- phenol (no reaction occurs)
- 2,4,6-tribromophenol

22. How many signals do you expect to see in the ^{13}C spectrum of methylbenzene?

- 1
- 3

- 5
- 7

23. A polymer is produced from two different monomers: a diamine and a dicarboxylic acid respectively. Other than the polymer, what substance is produced during the polymerisation?

- NH_3
- H_2O
- $H_2NC(=O)H$
- $H_2C=O$

24. In a Friedel Crafts acylation, a catalyst (halogen carrier) is required to facilitate the formation of the electrophile that gets attacked by the aromatic ring. Which of the following is **not** suitable for this purpose?

- Fe
- $FeCl_3$
- $AlCl_3$
- Cl_2

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